

# Sourasish Ghosh

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## SUMMARY

Machine Learning Engineer with hands-on experience building computer vision, anomaly detection, retrieval-augmented generation (RAG), and data engineering systems. Experienced in developing and deploying end-to-end ML applications using Python, FastAPI, Kafka, Redis, PostgreSQL, and Docker. Strong background in machine learning, atmospheric data analysis, and production-oriented AI systems.

## INTERNSHIP EXPERIENCE

**Spaceborn** Feb 2026 – Apr 2026

*Machine Learning Engineer Intern*

- Developed an unsupervised anomaly detection system using **Isolation Forest** to identify abnormal patterns in high-frequency multi-dimensional sensor streams.
- Designed a low-latency streaming architecture using **Apache Kafka**, **Redis**, and **Avro** serialization to support real-time ML inference pipelines.

**Excelerate** Dec 2025 – Jan 2026

*Data Analyst Intern*

- Led a team-based analytics project on global hiring datasets covering candidate and workforce trends.
- Performed exploratory data analysis and visualization using **Python**, **Excel**, and **Power BI**.
- Identified geographic hiring patterns and yield-management insights across datasets spanning more than 200 countries.

## PROJECTS

**End-to-End QLoRA and RSLoRA LLM Fine-Tuning Pipeline on Unsloth** ([GitHub](#)) 2026

- Engineered 4GB VRAM PEFT Pipeline: Finetuned a **2.6B** parameter model (**gemma-2-2b-it**) on an entry-level laptop GPU by building a constraint-driven training loop that eliminated out-of-memory crashes.
- Halved GPU Memory Footprint: Integrated **4-bit NF4** quantization, 8-bit optimizer states, and gradient checkpointing to reduce the active memory footprint to 2.0 GB, maintaining a **50% VRAM** safety overhead.
- Quadrupled Gradient Update Scaling: Implemented Rank-Stabilized **LoRA** (rsLoRA) at rank 16 to increase the mathematical weight update scaling by **400%** (4×) over standard QLoRA, driving a significantly lower final training loss.

**Real-Time Financial Fraud Detection and Telemetry Platform** ([GitHub](#)) 2026

- Engineered a full-stack fraud telemetry platform using **FastAPI**, **XGBoost**, **Redis**, and **PostgreSQL** that executed transaction classification and explainable **SHAP** reasoning in under *5ms*.
- Achieved an exceptional **93.85% Recall** rate and a **0.9376 PR AUC** during model evaluation on a 1.3-million-row dataset, maximizing true positive fraud detection.
- Eliminated a critical 'stateless' API vulnerability by utilizing an in-memory Redis cache with a **30-day** TTL to execute rolling customer velocity lookups in under 1ms.

**E-Commerce Multimodal RAG System** ([GitHub](#)) 2026

- Built a multimodal retrieval-augmented generation platform supporting semantic search across product descriptions and images.
- Generated text and image embeddings using **OpenCLIP** and stored vectors in **Qdrant**.
- Developed a **FastAPI** backend with Gemini-powered response generation and request rate limiting using **SlowAPI**.

**Python-based Finance Backend System** ([GitHub](#)) 2026

- Developed a backend financial services platform using **FastAPI**, **SQLAlchemy**, and **PostgreSQL**.
- Implemented **Role-Based Access Control (RBAC)** and secure authentication workflows.
- Leveraged **Pydantic** models for validation, type safety, and API consistency.

## RESEARCH EXPERIENCE

**M.Sc. Dissertation: Hybrid Fuzzy Random Forest Weather Classification** ([Github](#)) 2025

- Data Engineering & Validation:** Processed 24 years of daily **ECMWF IFS** data across 6 core parameters; executed *Chi-square* and *T-tests* to prove **Lifting Condensation Level** (*t*: 16.78) and **Diurnal Temperature Range** (*t*: 14.18) are critical discriminators for extreme weather, effectively filtering noise.
- Hybrid ML Architecture:** Engineered a hybrid model combining **Fuzzy Set Theory** with **Random Forest** to manage sensor imprecision and non-linear interactions; utilized *max-min composition* and balanced class weighting to eliminate majority bias on highly imbalanced disaster data.
- Performance & Insights:** Achieved **93.22%** overall accuracy (**0.8490 macro precision**) and **100% precision/recall** (F1: 1.0) for Heat/Cold Waves; proved that mapping Relative Humidity alongside Wind Speed and DTR in a fuzzy matrix drastically reduces false-alarm rates.

## EDUCATION

| Degree / Program                                    | Institution            | Timeline    |
|-----------------------------------------------------|------------------------|-------------|
| M.Sc. in Atmospheric Sciences (CGPA: 8.155)         | University of Calcutta | 2023 – 2025 |
| B.Sc. in Data Science and Applications (CGPA: 7.73) | IIT Madras (Distance)  | 2023 – 2027 |
| B.Sc. in Physics (Honours) (CGPA: 7.72)             | Chandernagore College  | 2020 – 2023 |

TECHNICAL SKILLS

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**Languages:** Python, SQL, Java, Bash, Fortran,  $\text{\LaTeX}$   
**Machine Learning & AI:** PyTorch, TensorFlow, Scikit-learn, OpenCLIP, Computer Vision, NLP, LLMs, RAG, LangChain, LangGraph, LoRA, QLoRA  
**Data & Databases:** Qdrant, ChromaDB, PostgreSQL, Redis, Apache Kafka, PySpark, Pandas, NumPy  
**Cloud, DevOps & MLOps:** Azure, GCP, AWS, Docker, Kubernetes (K8s), MLflow, Git, GitHub Actions, CI/CD Pipelines, VMs.  
**Frameworks & Tools:** FastAPI, Flask, Grafana, Prometheus, PowerBI

CERTIFICATIONS

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- Microsoft Azure Introduction to Cloud Infrastructure (AZ-900 series) ([Link](#))
- Simplilearn: Intro to Cloud Computing ([Link](#))
- Simplilearn: Machine Learning using Python ([Link](#))